Jack Basinski

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Summary

Circuit designer with real fab hours. Cut DRAM cycle time 400 hr/week at Micron, lead a 14-person ISO 5 cleanroom team, and design across analog, RF, and digital. Now building AI-assisted analog design tools in a silicon photonics lab.

Education

University of Washington Expected June 2027
Bachelor of Science in Electrical and Computer Engineering *Seattle, WA*

- GPA: 3.7, College Honors student, Dean's List scholar
- **Relevant Coursework**: Analog + RF IC Design, Photonics, Antennas, Computer Architecture, Digital Design

Experience

Photolithography Process Engineer Intern June 2026 – September 2026
Micron Technology *Boise, ID*

- Analyzed SPC and manufacturing data to track process variation and characterize DRAM photolithography steps
- Consolidated a two-step SOG/SOC coating flow into a single MLR process, cutting cycle time by 400 hr/week and freeing coat-tool capacity across the DRAM line

Lead Undergraduate Research Assistant May 2025 – Present
Washington Nanofabrication Facility *Seattle, WA*

- Coordinate ISO 5 cleanroom operations with 14 undergraduates, reporting to WNF staff and training new users
- Run semiconductor fabrication processes, including lithography, etch, and metrology, for 150+ academic and commercial facility users
- Characterized photoresist spin curves to quantify thickness variation for process qualification

Undergraduate Researcher October 2026 – Present
UW EMIT Lab (Prof. Sajjad Moazeni) *Seattle, WA*

- Develop a custom MCP server that lets Claude drive Cadence Virtuoso for AI-assisted analog IC design in a silicon photonics and co-packaged optics lab
- Refine run-to-run consistency so agent-generated designs pass parasitic extraction and reliability checks

Projects

CMOS 5T Operational Transconductance Amplifier | *Cadence Virtuoso* May 2026 – June 2026

- Sized, simulated, and optimized the amplifier using sensitivity analysis and transistor-level optimization
- Achieved 22.2 dB gain, 1.40 GHz GBW, and 108 MHz bandwidth while meeting stability requirements

Adaptive Interference Cancellation System | *Python, Arduino, LTSpice* January 2026 – March 2026

- Designed a Tow-Thomas biquad notch filter whose center frequency is retuned in real time by digital potentiometers, driven by FFT-based frequency detection
- Suppressed interference across the 2–20 kHz band with sub-kHz tuning accuracy

4×1 Microstrip Patch Antenna Array | *Ansys HFSS, Altium Designer* January 2026 – March 2026

- Simulated, fabricated, and measured a 5.8 GHz 4×1 microstrip patch array on FR-4, exceeding all specifications with 632 MHz measured bandwidth and 9.18 dBi peak gain
- Extended the design with a stacked parasitic-patch configuration, doubling measured bandwidth to 1.3 GHz and raising simulated peak gain to 11.1 dBi

Pipelined CPU | *SystemVerilog, Assembly, ModelSim* June 2025 – August 2025

- Architected a 5-stage pipelined CPU on ARM's LEGv8 instruction set, validating a 2.7× speedup over a non-pipelined implementation across benchmark algorithms
- Resolved data hazards with forwarding hardware, improving CPI by 70%

Technical Skills

Languages + AI Tools: Python, SystemVerilog, C, MATLAB, Assembly, Claude Code, MCP servers
Design + Test: Cadence Virtuoso + Spectre, LTspice, ModelSim, Ansys HFSS, Altium Designer, FPGA development, SolidWorks, oscilloscopes, VNA, soldering
Fabrication: Direct-write + contact lithography, evaporation, sputtering, CVD, RIE/DRIE, metrology, JMP
General Knowledge + Interests: Mandarin Chinese, Photography, Hiking, 3D Printing, Pearl Jam